RAHAL NANAYAKKARA

PhD student at the Cyber-Physical Systems Lab, University of California, Los Angeles
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EDUCATION

University of California, Los Angeles (UCLA)
PhD in Electrical and Computer Engineering — *Sep 2025 - Present*
MS in Electrical and Computer Engineering — *Sep 2023 - Jun 2025*
Advisor : Prof. Paulo Tabuada — **GPA: 4.00/4.00**

University of Peradeniya (UoP) - Sri Lanka — *2018 - 2023*
BSc. in Electrical and Electronic Engineering (Top of Class) — **GPA: 4.00/4.00**

Selected Classes (UCLA) : Linear Dynamic Systems, Nonlinear Dynamic Systems, Optimal Control, Geometric Nonlinear Control, Convex Optimization, Stochastic Systems, Neural Networks

EXPERIENCE

University of California, Los Angeles - Graduate Student Researcher — *Sep 2023 - Present*
Worked on several projects in areas of estimation, nonlinear systems and safety-critical control.

- *Robust Safety Critical Control* - Currently working on developing algorithms for synthesizing safety critical controllers in the presence of state uncertainty using modified control barrier functions. (In collaboration with AMBER Lab at Caltech, Pasadena)
- *State Estimation* - Worked on a computationally light, nonlinear state estimation algorithm that provides online error bounds from sampled output data. [repo]
- *Other Projects* - Worked on combining Neural Networks with a LiDAR localization algorithm that has provable error bounds. Also worked on using control theory to model the spread of epidemics on networks. Implemented Model Predictive Controller for autonomous cars for waypoint tracking in simulation environment.

Amazon, Prime Air - Applied Scientist (L5) Intern — *Jun 2025 - Sep 2025*

- Worked on the GNC Team of Amazon Prime Air. Implemented multipath mitigation techniques for improving positioning accuracy in GNSS degraded environments.

University of Peradeniya - Research Assistant — *Mar 2023 - Aug 2023*

- Worked on a biomedical engineering project for fetal health monitoring based on sensory data from abdomen mounted inertial sensors.

Vega Innovations - Electronic Engineering Intern — *Aug 2022 - Nov 2022*

- Worked on flux weakening control for high speed operation of Permanent Magnet Synchronous Motors in electric vehicles.
- Implemented over the air firmware updates on embedded microcontrollers.

PUBLICATIONS

Journal Papers

1. **R. Nanayakkara**, A.D. Ames and P. Tabuada, "Safety under state uncertainty: Robustifying control barrier functions," *Automatica* (2026).
2. X. Tan*, **R. Nanayakkara***, P. Tabuada and A.D. Ames, "A Duality-Based Optimization Formulation of Safe Control Design with State Uncertainties," *IEEE Control Systems Letters* (2026).

3. E. Daş, **R. Nanayakkara**, X. Tan, R.M. Bena, J.W. Burdick, P. Tabuada and A.D. Ames, "Safe navigation under state uncertainty: Online adaptation for robust control barrier functions," *IEEE Robotics and Automation Letters* (2026).

Conference Papers

1. **R. Nanayakkara**, A.D. Ames and P. Tabuada, "A Recursive CBF Framework for Safety under State Uncertainty," 2026 *IEEE 65th Conference on Decision and Control (CDC).*
2. **R. Nanayakkara** and P. Tabuada, "Predicting Viral Evolution from a Single Early Measurement Using the Target Cell Limited Model," 2026 *IEEE 65th Conference on Decision and Control (CDC).*
3. J.P. Silvestre, **R. Nanayakkara** and P. Tabuada, "Nonlinear Observers with Tighter Online Error Bounds," 2024 *IEEE 63rd Conference on Decision and Control (CDC).*
4. L.B.I.P. Thilakasiri, D.M.P.M. Alwis, **R.T. Nanayakkara**, G.M.R.I. Godaliyadda, M.P.B. Ekanayake and J. V. Wijayakulasooriya, "Fetal Movement Identification Using Spectrograms with Attention Aided Models and Identifying a Set of Correlating Parameters with Gestational Age," *2023 IEEE 17th International Conference on Industrial and Information Systems (ICIIS).*
5. L.B.I.P. Thilakasiri, D.M.P.M. Alwis, **R.T. Nanayakkara**, G.M.R.I. Godaliyadda, M.P.B. Ekanayake and H.M.V.R. Herath, "Integrated Video Based Crowdedness Forecasting Framework with a Review of Crowd Counting Models," *2021 IEEE 16th International Conference on Industrial and Information Systems (ICIIS).*

Workshop Presentations

1. **Rahal Nanayakkara**, Ersin Das, Xiao Tan, Aaron Ames and Paulo Tabuada. "Safety Under State Uncertainty: Robustifying Control Barrier Functions," *45th Southern California Control Workshop 2025.*

SKILLS

Programming Languages	Python, MATLAB, Java, C/C++
Software Tools	Git, Numpy, PyTorch, ROS, OpenCV
Hardware Skills	PCB design & development, wiring harness construction, SMD soldering
Technical Knowledge	Linear and Nonlinear Control Theory, Optimization, Machine Learning

AWARDS

1. **C.A. Hewavitharana prize for the best performance in engineering, 2023**: Awarded for overall best academic performance in the school of engineering, University of Peradeniya.
2. **CEB Gold medal for best performance in electrical & electronic engineering, 2023**: Awarded to top of the class in the Department of Electrical and Electronic Engineering.
3. **Prof J.A. Gunawardena prize for the best performance in control engineering, 2023**: Awarded for best academic performance in controls related subjects (s-domain control, discrete and continuous time state space based control, basic nonlinear control).
4. **Bartholomeusz Prize for Engineering Mathematics, 2020 & 2021**: Awarded for best performance in mathematics modules in the 1st and 2nd year of the undergraduate program.
5. **Bronze Medal at the Sri Lanka Mathematics Olympiad, 2017**: Ranked 5th nationally, with High Distinction. Conducted by the Sri Lanka Olympiad Mathematics Foundation.

SELECTED ACHIEVEMENTS

IEEEXtreme 14.0 - 24 hour global algorithmic programming competition *2020*
National Rank - 2, World Rank - 68 out of 4,132

IEEEXtreme 13.0 - 24 hour global algorithmic programming competition *2019*
National Rank - 5, World Rank - 130 out of 4,134

MoraXtreme 4.0 - 12 hour algorithmic programming competition *2019*
Placed 1st out of 150 teams in Sri Lanka

SLIIT Robofest - Autonomous maze navigation (Micromouse) robotics competition *2019*
Placed 3rd out of 50 teams

G.C.E.(A/L) - Competitive national examination for university entrance *2016*
District Rank - 1 out of 2,987. National Rank - 14 out of 33,608. Z - score - 2.7522

SELECTED PROJECTS

LiDAR localization UCLA, 2025
Performed comparison of LiDAR localization algorithms based on experimental data collected in the Lab. Combined provable error bounds from a previously developed LiDAR localization algorithm with Neural Network based localization algorithms. (ROS, Python)

Model Predictive Controller (MPC) for autonomous cars UCLA, 2024
Implemented an MPC controller for waypoint tracking in an autonomous vehicle simulator. Implementation was done through dynamic linearization along the trajectory. (Python)

Four Degree of Freedom Robotic Arm UoP, 2023
Fabricated a miniature 4 DOF robotic arm from acrylic glass. Implemented pick and place motions using forward and reverse kinematics for trajectory calculation. (C++)

CAN bus based LCD for vehicle engine monitoring UoP, 2022
Designed and assembled an LCD display to monitor engine parameters using CAN data. Used an ESP32 microcontroller for CAN decoding and LCD control. (C++)

Autonomous Maze Navigating Robot (Micromouse) UoP, 2019
Designed and built custom hardware (including sensors) for an autonomous maze navigating robot. Implemented low level controllers and algorithms for navigation and motion planning. (C++)

SERVICES

Reviewer Roles

- Served as a reviewer for Automatica (2026), IEEE Control Systems Letters (2025, 2026), IEEE Robotics and Automation Letters (2026).
- Served as a reviewer for IEEE International Conference on Robotics and Automation (2027), IEEE Conference on Decision and Control (2026), IEEE International Conference on Industrial and Information Systems (2025).

Volunteering

- Conducted "Introduction to python" webinars for school students and first year undergraduates hosted by the University of Peradeniya, Sri Lanka. (2021)
- Conducted Arduino and IOT workshops for school students and teachers to promote IT education among young students in Sri Lanka. (2019 & 2021)